

Instructions for the Activity #1

MECH3424 measurement and instrumentation

(Virtual) Statistical analysis of amplifier characteristic(s)

Aim: The aim of this laboratory is to :

- 1) Be able to read and correctly interpret information on a datasheet
- 2) Appreciate the importance of statistical distributions in components.
- 3) Use numerical methods to explain and compare with the data in datasheets

Purpose: When designing any instrument, you may want to use a part within that instrument with a specific characteristic, but in practice, the part will have a random distribution of values. This lab attempts to have you review and analyse several characteristics of some operational amplifiers, comparing that data to the manufacturers datasheet. Your ability to read the manufacturers datasheet should be improved through the process of statistical analysis and reflection of the data analysis. You will gain this understanding by **NUMERICALLY** simulating a large number of parameters. This simulation would represent the data measured by a company actually making and measuring all these parameters, but the simulation process is much easier than making and measuring, allowing analysis and data mining to be completed.

The fundamental process to perform this lab requires:

- 1) The detailed analysis and comparison of the voltage and current data from the manufacturers specifications using statistical techniques
- 2) The use of numerical statistical models based on **python** to better understand what the datasheets are telling you
- 3) Consideration and assumptions of non-normal distributions, symmetric and one-sided distributions used when analysing data and their impact
- 4) Presenting results in a high quality professional manner

NOTE: You need to create the basic code for the analysis and not use complex existing packages. Only a basic python3 installation along with numpy and matplotlib packages can be used– the use of any other packages within your code is not permitted. These packages are all that is required to gain the understanding required.

Preparation before Lab #1

- Please review the MECH4424/ MECH3424 notes:
 - o Chapter 3: Signal Conditioning
 - o Chapter 4: Probability & statistics – this chapter is particularly important so read ahead
 - o Chapter 5: Data Presentation – required for the preparation of the report
- Please all review the datasheets provided on see LMS. Review the overall datasheet for each op-amp and how it specifies (for the particular op-amp) the input offset voltage, input offset current, bias current and temperature dependence specification for each amplifier. Review the course reader to understand how these parameters are reported in datasheets.
- Please review the Laboratory FAQ on LMS. Every year students ask questions and these questions are then posted and answers on the LMS FAQ. Please check there first before emailing the unit coordinator with questions about the laboratory.

Method to undertake the (virtual) simulation:

Introduction: In order to statistically analyse the data, you will need to create a **simulation** to generate a large number (10^5) of random values which represent a distribution of:

- Input offset voltage
- Input offset current & bias current
- Input offset voltage drift with temperature

In your report, for each dataset, you will need to create a histogram with appropriate bin size (some research/investigation may be required), where the data is presented and analysed using the skills developed in this unit. The analysis required for each parameter is discussed below along with some background information (shown in the shaded region)

Study#1 Background: op-amp offset voltage

For this analysis consider **the OPA445AP datasheet**. The input offset voltage is defined below in Figure 1 as (V_{io}) and it is typically the result of a zero-mean symmetric random distribution described by a standard deviation σ .

Input Offset voltage (V_{io}): Voltage across inputs required to give zero output voltage.

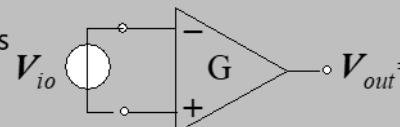


Figure 1: Definition and illustration of how the input offset voltage for an operational amplifier is defined.

Study #1a: Input offset voltage

You are required to create a simulation to generate a large number (10^5) of random values to create a distribution of input offset voltage values designed to match the datasheet data for **op-amp OPA445AP**. You will need to understand the values reported in the datasheet and how these relate to the values which can be used in a statistical model. Tasks:

- Analyse the mean, standard deviation and max/min for the dataset and compare to the datasheet values and discuss any differences
- Build a histogram for the dataset size which would represent the percentage of occurrences of the samples within each bin of input offset voltage on the histogram.
- Compare the histogram simulated to the histogram given in the datasheet in a meaningful manner. Explain the differences.

Study #1b: Sorting Analysis:

Sorting parts is a way to improve quality and charge more for those higher quality items. Extract samples from the offset voltage dataset which would allow you to specify in a datasheet that you have op-amps for sale with a typical value of 1 mV and a maximum of 3 mV (see *OPA445BM*).

- Explain how the *original* dataset was sorted to achieve this goal and how this would be done in a manufacturing environment
- Explain how you were able to get as close as possible to the desired typical value of 1 mV and max 3mV for your simulated dataset.
- Summarise :
 - the statistics and the number of the op-amps that you could sell with this higher performance typical input offset of 1 mV and max 3mV

- o the statistics and the number of the op-amps that remain after these higher quality op-amps are extracted
- Discuss the higher-quality (typical input offset voltage of 1 mV and max 3mV) and the remaining op-amps (lower quality) with reference to the specifications for the *OPA445BM* and *OPA445AP*
- What do the histograms of the two datasets look like ?

Study#2 Background: Input offset current and bias current analysis

Input offset current (I_{io}) and bias currents (I_{ib}) are defined from a **physical parameter** I_{ib1} and I_{ib2} , which is the current flowing into or out of the op-amp and are defined as:

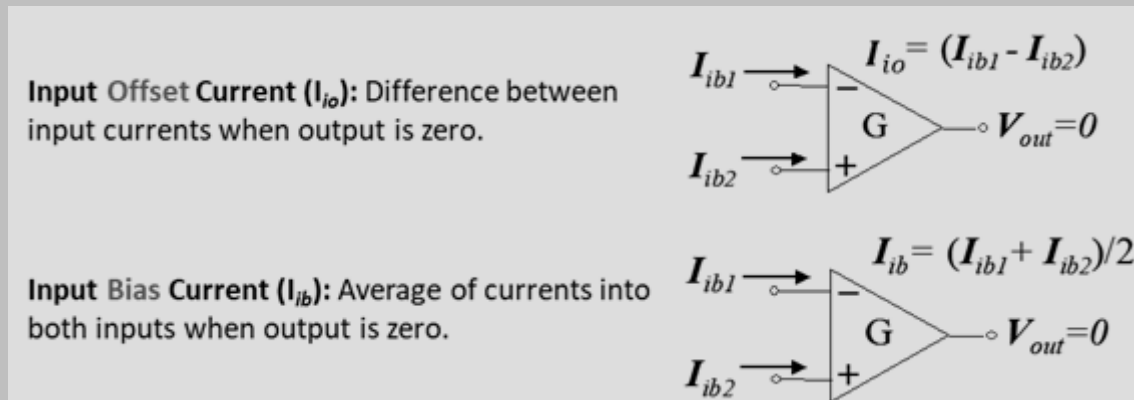


Figure 2: Definition and illustration of how the (a) the input offset current and (b) the input bias current, for an operational amplifier is defined.

Importantly, these **physical parameters** I_{ib1} and I_{ib2} represent either flow into (or out of) a transistor connected to the front of the op amp as illustrated in Figure 3a and 3b. For the LMS358 shown in Figure 3c, the input uses PNP transistors and hence the physical parameter I_{ib1} and I_{ib2} represents only negative (flowing out) current flow – it can't be positive. As such models that assume a symmetric distribution for the current are not useful to explain this data. In this exercise, consider **the LM358 datasheet**, and assume that the **physical parameter** I_{ib1} and I_{ib2} are each is described by an unspecified (one sided) random distribution. **The two inputs are nearly identical but not perfectly matched.** Currents I_{B1} and I_{B2} come from the same chip and are designed to be the same, so they're strongly linked. A chip with a high leak on one input tends to have a high leak on the other. But manufacturing is never perfect, so there's a small difference between them.

Use the simulated random values for I_{B1} and I_{B2} to calculate the input offset current (I_{io}) and bias current (I_{ib}) using the equations defined in Figure 2 above. In interpreting these two currents remember that:

- $I_{ib} = (I_{B1} + I_{B2})/2$ so that if I_{B1} and I_{B2} are both negative then I_{ib} is negative
- $I_{io} = (I_{B1} - I_{B2})$ so that given I_{B1} and I_{B2} are both highly correlated but still random, then values of I_{io} are just as statistically likely to be positive or negative resulting in a symmetric distribution for I_{io} .

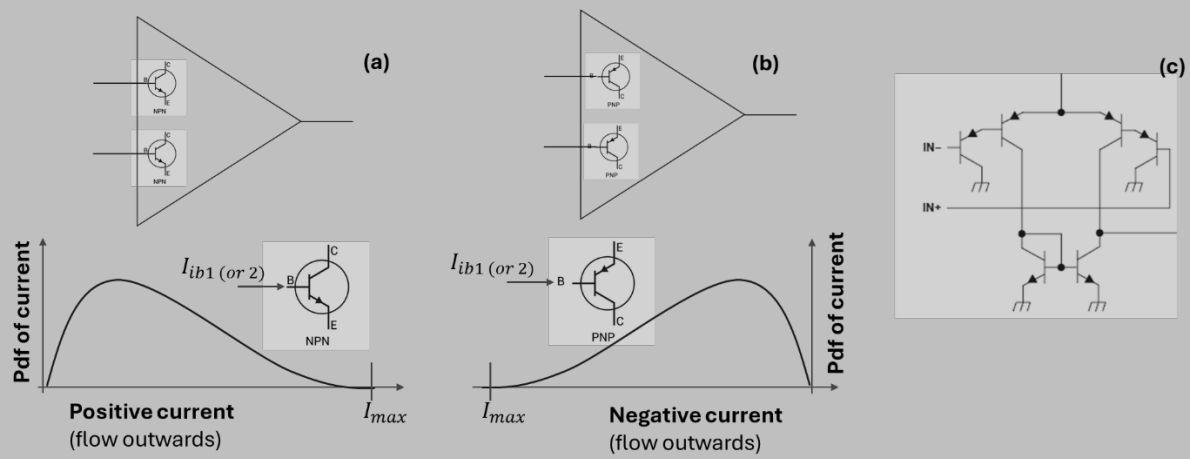


Figure 3: Illustration of the expected distribution of current flowing (a) into or (b) out of op amps with different input transistors. (c) extract from LM358 data sheet showing it is based on input PNP transistors (hence current flow is out).

Even though I_{B1} and I_{B2} are both strictly negative and nearly equal, their **difference** (I_{io}) should be **symmetric and centred on zero** — because the shared part (the part common to both called the common mode) cancels out. This is a good demonstration of how operations on distributions (adding, subtracting, multiplication) produce new distributions with different shapes. As a result the distributions for I_{B1} and I_{B2} can be formed from a common ($c(I)$) and differential distribution ($d(I)$) by any number of forms including:

Form #1 (Addition)

$$I_{B1} = c(I) + \frac{d(I)}{2}$$

$$I_{B2} = c(I) - \frac{d(I)}{2}$$

Form #2 (multiplication)

$$I_{B1} = c(I) \left(1 + \frac{d(I)}{2} \right) \quad (\text{Eq 1})$$

$$I_{B2} = c(I) \left(1 - \frac{d(I)}{2} \right) \quad (\text{Eq 2})$$

This is because the common distribution of values, $c(I)$, represents the leakage current both inputs have in common while the differential distributions, $d(I)$, represented the small difference between the two inputs. Because both inputs are built from the same **common distributions**, they come out correlated automatically. You never apply a correlation technique. You only arithmetically *calculate* the correlation at the end to confirm it worked.

Study #2: Input offset current and bias current analysis

The key idea that makes this study challenging is that you will **not** model I_{ib} and I_{io} as separate things. The currents I_{ib} and I_{io} are *not independent* but are both built from the same two underlying currents, I_{B1} and I_{B2} . So:

1. You model distributions of the two underlying currents I_{B1} and I_{B2} as your starting point.
2. You then **calculate** I_{ib} and I_{io} from them using the two formulas :

$$I_{ib} = (I_{B1} + I_{B2})/2 \quad (\text{Eq 3})$$

$$I_{io} = (I_{B1} - I_{B2}) \quad (\text{Eq 4})$$

Distributions: Chose the correct shaped distribution based on the physical description of the op-amp. Think about the distributions:

- Common distribution (a leak current): **one-side or 2 sided ?**
- differential (either input could be larger): **one-side or 2 sided ?**

A note on the simulation. When using python for this task the only imports allowed are **numpy** and **matplotlib** only. **No other imports are permitted** . Build distributions from NumPy's statistical generators or using the random number generation techniques covered towards the end of Chapter 4. Explore any methods to generate the distributions required. **Importantly**, this problem does not need covariance matrices, copulas, Cholesky decompositions or other advanced correlation techniques. Don't try to use such correlation techniques to generate the distribution. If two quantities are made from a **common distribution** as discussed in the **Study#2 Background**, they are automatically correlated. You need to appreciate (and show) that having a common distribution forces the data to be correlated. You must *verify* that the random samples you have generated are correlated but only after forming the two datasets I_{B1} and I_{B2} . Correlation is achieved by **construction** because you used a shared (common) distribution.

Hints:

1. **The distributions are the *physical* quantities, not the datasheet quantities.** You need to call a random number generator for the common distribution and the differential distribution. *Everything else is arithmetic.* Do not try to create complex cross-correlated distributions.
2. I_{ib} and I_{io} are the things to generated from the other distributions. Do not generate I_{ib} and I_{io} directly as doing so will require complex correlation techniques.

Once you have chosen the shared and common distributions and chosen the form to combine them (examples in Equ 1 and Equ 2), you can use the distributions to generate I_{ib} and I_{io} given in Eq 3 and Equ 4 above. From the calculated values of I_{ib} and I_{io} determine how the statistical parameters (such as μ , σ , max/min and any others) of the datasets and compare to the typical and max/min values of the input offset current (I_{io}) and bias currents (I_{ib}) in the datasheets. From this comparison, determine if you have correctly matched the data for input offset current (I_{io}) and bias currents (I_{ib}) using your distributions and parameters used in the distribution. You may need to evaluate a number of different distributions and parameters to determine the most effective one to explain the data. If you need to generate a custom one-sided or symmetric distributions, consider using the random number generation techniques covered towards the end of Chapter 4. You need to demonstrate (investigate) the correlation between IB1 and IB3, which can be done for example using (for example) `np.corrcoef(IB1, IB2)[0,1]`. Determine what parameters affect this correlation and discuss.

Study #3: Temperature dependence - input offset voltage drift

For this analysis return to **the OPA445AP datasheet**.

Review the input offset voltage drift production distribution for the **Temperature dependence of offset voltage in the OPA445AP datasheet**. Fit a function or piecewise shape that best fits this distribution. Using this function or piecewise fit, you need to create a random number generator

that will produce the required number of samples having this distribution, using the methods discussed in this unit (specifically chapter 4)

You are required to create a simulation to generate a random distribution of input offset voltage **drift values** based on the offset voltage drift production distribution in the datasheet for **the op-amp OPA445AP**. Simulate the required samples for the temperature dependence of offset voltage. Analyse the data to:

- Create summary statistics of the data
- Compare in a meaningful way, the simulation to the data provided in the datasheet for **the op-amp OPA445AP**
- Calculate datasets for input offset voltage at the minimum and maximum operational temperature of the op-amp and compare the datasets. Explain your process.
- Summarise the statistics at the **max/min operational temperature** for the simulated input offset voltage datasets and compare to the datasheet histogram, typical and max/min values reports for input offset voltage. Does it significantly change the typical and max/min values.

Resources

Some resources which might be useful if you are using python:

Introduction to simulating random data using numpy in python:

<https://soumenatta.medium.com/simulating-randomness-using-pythons-numpy-random-module-ad96023daee7>

How to generated random number in python (not only 2nd ½ of webpage is relevant as it relates to use of numpy – you cannot use the import random package for this assignment)

<https://machinelearningmastery.com/how-to-generate-random-numbers-in-python/> ,

Outcome: At the end of this laboratory, students should be able to (through discussion with the unit coordinator, with team members, with other teams and/or in your report):

- Understand how the manufacturers obtain the data (typical/min/max) that is documented in the datasheets for the input offset voltage and current
- Understand how to create simulated datasets which can be used to represent manufactured data
- Explain the quantities simulated and the statistical properties of interest
- Present measurements in a suitable format for the reader to appreciate the simulations and include a discussion of the findings.
- Analyse the simulations taken including any limitations, assumptions or irregularities
- Analyse the statistical distribution of the data and the summary statistics
- Provide a comparison of the results of the simulations and with reference to other source of information